Small stories around a bike

How a Bike Chain Works

Name _____ Date _____

Use numbers or a drawing to share your thinking.

1. Little joins 28 paper links, then adds 17 more. How many links are in the paper chain?

28 + 17 = _____

Show your thinking.

2. Big has 64 beads and threads 26 onto a model chain. How many beads are left?

64 − 26 = _____

Show your thinking.

3. Little records 23 slow pedal turns, then 19 more. How many pedal turns are recorded?

23 + 19 = _____

Show your thinking.

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4. A paper gear has 52 drawn teeth. Little colors 18. How many teeth still need color?

52 − 18 = _____

Show your thinking.

5. A bike has 32 teeth on its front sprocket and 16 on its back sprocket. How many teeth is that in all?

32 + 16 = _____

Show your thinking.

6. Little plans 15 gentle driveway laps and finishes 7. How many laps are left?

15 − 7 = _____

Show your thinking.

A push has a path

The pedals were far from the back wheel. A loop of small metal links joined them. Little wondered how a push could travel.

Bend at each pin

Tiny pins join the links. The chain bends at these joints without stretching like rubber. Sprocket teeth hook into the chain as it moves.

Turning, then pulling

Pedals turn the front sprocket. Its teeth pull the chain. The chain pulls the back sprocket, which turns the back wheel.

A loop on the table

Big turned a cardboard wheel. It pulled the bead loop, and the loop turned the smaller wheel. Little followed one bead around.

One turn becomes two

The front had 32 teeth. The back had 16. One pedal turn carried enough chain past the back sprocket to turn it twice.

A push goes with you

Little pedaled beside Big. The chain carried each pull to the back wheel. Where else could a loop carry motion?

Make a little book

1 Fold it into 8 boxes. Open it flat.

2 Fold short end to short end. Cut the thick line. Open it.

3 Fold the long way, pictures out. Push the ends in.

4 Fold into a book. Page 1 goes on the front!

WONDER DAILY

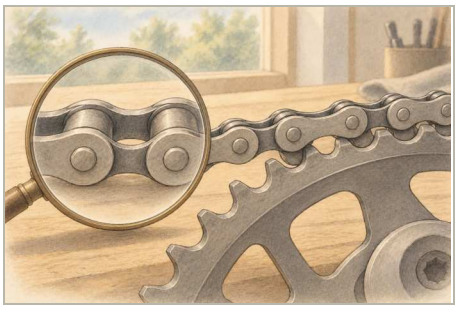
How a Bike Chain Works

Give the chain a clear command

How a Bike Chain Works · Curiosity day

Name _____ Date _____

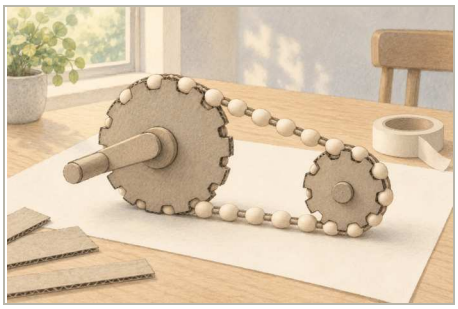
Choose a still view and tell what to count or draw.



Bend at each pin



Teeth to count



A loop on the table

Draw two sprockets joined by a chain. Add an arrow along the top of the chain.

chain · link · pin · sprocket · teeth · pedal · gear · pull · turn · wheel

Trace … name the loop you will follow

Count … name the teeth on a still sprocket

Draw … show how a loop joins two wheels

Does your command begin with an action word, such as Trace, Count, or Draw?

How a Bike Chain Works

01 How can metal bend around a wheel?



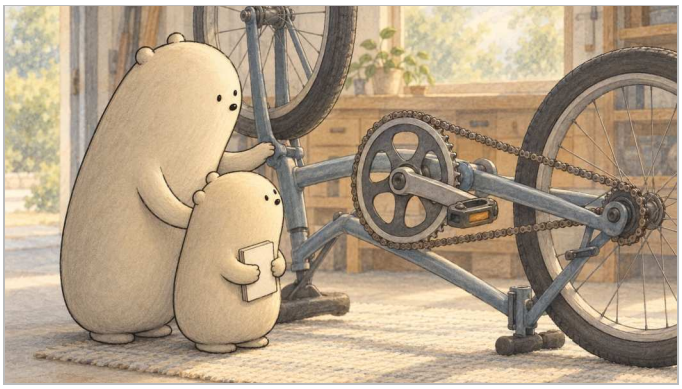
Pins let links bend; teeth catch the chain.

Look closer. Find the little round pins between links.

A little explanation. A chain is a loop of links joined by tiny pins. Each joint lets neighboring links turn. The links stay the same length while the whole loop bends around a sprocket.

Try this. Draw three links joined by dots for pins. Bend your drawn path around a circle.

02 How does the pedal reach the back wheel?



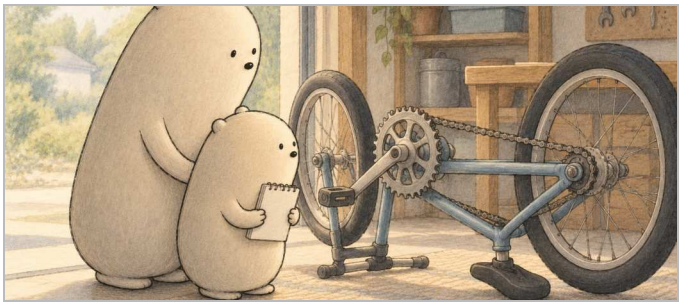
The chain connects the pedal's sprocket to the back wheel's sprocket.

Look closer. Follow the top of the chain between the two sprockets.

A little explanation. A sprocket is a toothed wheel. The front one, called the chainring, turns with the pedals. Its teeth pull the chain; the chain pulls the back sprocket's teeth. That sprocket turns the back wheel as you pedal forward.

Try this. Use the diorama to follow turning, pulling, then turning again.

03 What can a still bike show us?



Count teeth while every part is still.

Look closer. Look for the chain's links and the sprockets' teeth.

A little explanation. Teeth fit into the spaces between the chain's rollers. This fit lets the sprocket pull the chain without slipping. Count the front and back teeth while everything is still.

Try this. Count teeth first, then watch one slow pedal turn. Keep Big beside you. Count only while the bike is still. Big holds an upside-down bike steady on a mat; turn only the pedal slowly by hand, with fingers well away from chain, sprockets, and spokes whenever anything moves. Leave oiling and repairs to Big. At the table, try notched cardboard wheels and a bead loop, or tape paper strips into a chain. Use child scissors only with Big's yes.

04 Can one pedal turn make two wheel turns?



More front teeth can mean more back-wheel turns.

Look closer. Compare the big front sprocket with the small back one.

A little explanation. With 32 teeth in front and 16 behind, one pedal turn gives two back-wheel turns. A big front and small back help with faster riding. A small front and big back make pedaling easier uphill. These choices are gears.

Try this. Draw one circle for a pedal turn and two circles for the back wheel's turns.